

Notes: Giroud, Liu, and Mueller (JFE, 2026)

Innovation Spillovers Across U.S. Tech Clusters

Contents

1	Big picture	3
2	Data, units, and key variables	3
2.1	Unit of analysis and sample construction	3
2.2	What is a “cluster”?	3
2.3	Outcome variables	3
2.4	Cluster size and connected clusters	4
3	Empirical models (reduced-form regressions)	4
3.1	Within-cluster spillovers	4
3.2	Cross-cluster spillovers	4
3.3	Interpretation of β	5
4	Main empirical results (table-by-table interpretation)	5
4.1	Table 1: summary of main patterns	5
4.2	Table 2: descriptive statistics	5
4.3	Table 3: within-cluster spillovers (Eq. 1)	7
4.4	Table 4: cross-cluster spillovers (Eq. 2)	7
4.5	Table 5: non-innovating plants and “placebo” connected clusters	8
4.6	Table 6: instrumental-variable (2SLS) estimation	8
4.7	Table 7: geographic distance and the lack of decay	9
4.8	Table 8: patent citations as a knowledge-flow mechanism	9
4.9	Tables 9–11: wedge dispersion and policy relevance	9
5	Theoretical framework: model, welfare, and the innovation wedge	10
5.1	Setup and preferences (Eqs. (3)–(4))	10
5.2	CES demand, pricing, and profits (Eqs. (5)–(8))	11
5.3	Market access and plant revenues (Eqs. (6), (9), (13))	11
5.4	Plants, firms, and knowledge (Eqs. (10)–(11))	11
5.5	Plant profits and the firm problem (Eqs. (12), (14)–(16))	12
5.6	Uniqueness (Lemma 1)	12

5.7	Welfare, private value, social value (Eqs. (18)–(24))	13
5.8	Optimal subsidy (Proposition 1) and special case (Eq. (26))	14
5.9	Location-level wedge (Proposition 2) and network interpretation	14
6	Taking the wedge to the data: calibration logic behind Tables 9–11	15
7	Conclusion	15
7.1	Summary	15
7.2	Economic intuition: why connected clusters generate distant spillovers	16
7.3	What to remember (for research design)	17
7.4	Takeaway	17

1 Big picture

The paper studies *innovation spillovers* across U.S. tech clusters and shows that these spillovers propagate through *multi-location firms*. Empirically, the authors identify:

- **Within-cluster spillovers:** plants innovate more and become more productive when the local cluster (city $\times$ field) becomes larger.
- **Cross-cluster spillovers:** plants also benefit from innovation taking place in *other* clusters *connected* to them through their parent firm’s network of innovating plants.
- **A key distance fact:** cross-cluster spillovers do *not* decay with geographic distance (consistent with within-firm diffusion channels such as internal communication).

On the theory side, the paper builds a multi-location model with within-location and cross-location knowledge spillovers and derives a *sufficient statistic*—the **social–private innovation wedge** γ_n/β_n —that measures how much a location under-invests in innovation relative to the social optimum. The wedge is larger in more interconnected clusters, implying policy should account for the *network position* of clusters, not only their size.

2 Data, units, and key variables

2.1 Unit of analysis and sample construction

The main unit is a **plant-year** (Census plant data matched to USPTO patent data). The sample covers 1976–2018 and focuses on *innovating plants* (plants with inventors) for innovation outcomes; non-innovating plants are used for placebo and mechanism tests.

2.2 What is a “cluster”?

A cluster is defined at the level of:

$$\text{Cluster} \equiv (\text{city} \times \text{research field}).$$

- **Cities:** BEA economic areas.
- **Fields:** 1-digit CPC codes (broad patent technology fields).

2.3 Outcome variables

Let i index plants, c cities, f fields, and t years.

- **Inventor productivity (plant-level):**

$$y_{icft} = \log \left(\frac{\text{patents}_{icft}}{\text{inventors}_{icft}} \right).$$

This is defined only for innovating plants.

- **Plant-level productivity:** revenue-based TFP (TFPR), measured as the residual from a plant-level production function regression of output on capital, labor, and materials (in logs), estimated separately by 3-digit SIC industry $\times$ year.

2.4 Cluster size and connected clusters

Within-cluster size. For plant i in city c and field f at time t , define

$$\text{ClusterSize}_{icft}^{\text{within}} \equiv \log(\#\{\text{inventors at other firms in } (c, f, t)\}).$$

Connected clusters and cross-cluster size. A plant's parent firm may have innovating plants (with inventors) in *other* cities in the same field f . Those other city-field clusters are the plant's **connected clusters**. Define

$$\text{ClusterSize}_{icft}^{\text{cross}} \equiv \log \left(\sum_{\substack{c' \neq c: \\ \text{firm has innovating plants in } (c', f, t)}} \#\{\text{inventors at other firms in } (c', f, t)\} \right).$$

Crucially, connected clusters *exclude* other cities where the firm has plants but *no* inventors.

3 Empirical models (reduced-form regressions)

3.1 Within-cluster spillovers

The within-cluster specification is:

$$y_{icft} = \gamma_i + \gamma_{ct} + \gamma_{ft} + \gamma_{cf} + \beta \text{ClusterSize}_{icft}^{\text{within}} + \varepsilon_{icft}. \quad (1)$$

Fixed effects absorb:

- γ_i : time-invariant plant heterogeneity,
- γ_{ct} : city-year shocks,
- γ_{ft} : field-year shocks,
- γ_{cf} : time-invariant city-field comparative advantage.

Identification comes from within-plant variation in local cluster size, conditional on these fixed effects.

3.2 Cross-cluster spillovers

The cross-cluster specification replaces local cluster size with connected-cluster size:

$$y_{icft} = \gamma_i + \gamma_{cft} + \beta \text{ClusterSize}_{icft}^{\text{cross}} + \varepsilon_{icft}. \quad (2)$$

The key control is γ_{cft} (city $\times$ field $\times$ year fixed effects), which absorbs any *local* city-field-year shocks. Thus, β is identified by comparing plants in the *same* city-field-year that belong to different firms whose *connected clusters* differ in size.

3.3 Interpretation of β

Because both the dependent variable and cluster size are in logs, β is an **elasticity**. For example, $\beta = 0.021$ implies that a 10% increase in connected-cluster size raises inventor productivity by about 0.21%.

4 Main empirical results (table-by-table interpretation)

4.1 Table 1: summary of main patterns

Table 1 is an executive summary organized by:

- **Within-cluster** vs. **cross-cluster** spillovers,
- **Innovating** vs. **non-innovating** focal plants,
- whether **connected clusters contain inventors**.

The key message is an **inventor-based mechanism**:

1. Within-cluster spillovers operate for innovating plants (innovation and TFP rise), but not for non-innovating plants.
2. Cross-cluster spillovers require that the connected clusters contain inventors; if connected clusters contain only non-innovating plants, spillovers vanish.
3. Cross-cluster spillovers can raise TFP even at non-innovating focal plants, consistent with knowledge flowing within the firm and improving production efficiency.

4.2 Table 2: descriptive statistics

Table 2 provides basic magnitudes for the matched plant–patent sample. A typical plant has about 700 employees and substantial shipments; the average plant has about 5 inventors and produces about 11 patents per year (in its field). The average plant is connected to about 3 other clusters in its field, and the total number of *other-firm* inventors across those connected clusters is large, indicating economically meaningful variation in the explanatory variable. The firm-level panel shows that innovative activity is spread across multiple locations for many firms.

Table 1

Summary of main results.

Connected clusters with innovating (non-innovating) plants are clusters in other cities, in the focal plant's field, where the focal plant's parent firm has plants with (without) inventors. Innovating (non-innovating) plants are plants with (without) inventors. Log(pat/inv) is the ratio (in logs) of the number of patents to the number of inventors; this measure is only available for innovating plants. TFP is (revenue-based) total factor productivity (i.e., TFP). "+" indicates a (positive and statistically significant) spillover effect; "○" indicates no spillover effect.

Within-cluster spillover:		
	Log(pat/inv)	TFP
Innovating plant	+	+
Non-innovating plant	NA	○
Cross-cluster spillover:		
	Log(pat/inv)	TFP
Innovating plant; Connected clusters with innovating plants	+	+
Non-innovating plant; Connected clusters with innovating plants	NA	+
Innovating plant; Connected clusters with non-innovating plants	○	○
Non-innovating plant; Connected clusters with non-innovating plants	NA	○

Table 2

Descriptive statistics.

This table presents plant- and firm-level statistics for the main analysis sample consisting of 57,000 plant-year observations and 11,000 firm-year observations. Patents is the total number of patents (in the plant's field) per plant and year. Connected clusters is the number of other cities in which the firm has plants (in the plant's field) with inventors. Connected cluster size is the total number of other firms' inventors (in the plant's field) across all connected clusters. Cities are BEA economic areas. Fields are based on 1-digit CPC codes. Inventor share at the firm-plant level is the ratio of the firm's plants with inventors to all plants of the firm, averaged across all firms. Inventor share at the firm-county (city, state) level is the ratio of counties (cities, states) in which the firm has plants with inventors to all counties (cities, states) in which the firm has plants, averaged across all firms. The sample period is from 1976 to 2018.

	(1) Mean	(2) Std. dev.
Plants:		
Employees	699	1,623
Shipments (\$M)	275	1,670
Inventors	4.99	13.67
Patents	10.87	43.07
Connected clusters	3.22	2.81
Connected cluster size	1915	2,748
Firms:		
Employees	6853	13,280
Shipments (\$M)	2524	6,730
Plants	17.51	23.19
Counties	15.23	18.47
Cities	11.77	11.72
States	9.16	7.33
Inventor share:		
Plants	0.56	0.31
Counties	0.55	0.31
Cities	0.52	0.33
States	0.49	0.34

4.3 Table 3: within-cluster spillovers (Eq. 1)

Table 3 estimates Eq. (1).

- **Columns (1)–(4): innovating plants.** The coefficient on $\log(\text{cluster size})$ is positive and highly significant for both inventor productivity and TFP. The preferred (most saturated) fixed-effect specification is columns (3)–(4), which include plant FE and the full set of $\text{city} \times \text{year}$, $\text{field} \times \text{year}$, and $\text{city} \times \text{field}$ fixed effects.
- **Economic magnitude.** In column (3), the elasticity of $\log(\text{patents/inventor})$ with respect to local cluster size is about 0.076. In column (4), the elasticity of TFP is about 0.023. Thus, doubling local cluster size predicts about 7.6% higher inventor productivity and 2.3% higher TFPR for innovating plants.
- **Column (5): non-innovating plants (placebo).** The coefficient is small and statistically insignificant: local cluster size does not raise TFP when the plant has no inventors. This supports the view that local spillovers require internal absorptive capacity at the receiving plant.

Table 3

Within-cluster innovation spillovers

This table presents variants of the regression in Eq. (1) in which the dependent variable is either the ratio (in logs) of the number of patents to the number of inventors (all in the plant's field) at the plant level or plant-level TFP. Cluster size is the number of other firms' inventors in the plant's city and field. In column (5), the sample consists of plants without inventors. For these plants, fixed effects are based on the firm's main field, and cluster size is the total number of other firms' inventors in the plant's city in the firm's main field. Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Log(pat/inv) (1)	TFP (2)	Log(pat/inv) (3)	TFP (4)	TFP (5)
Log(cluster size)	0.078*** (0.010)	0.029*** (0.004)	0.076*** (0.013)	0.023*** (0.007)	0.003 (0.003)
Field FE	Yes	Yes	–	–	–
Year FE	Yes	Yes	–	–	–
Field $\times$ Year FE	No	No	Yes	Yes	Yes
City $\times$ Year FE	No	No	Yes	Yes	Yes
City $\times$ Field FE	No	No	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes	Yes
Observations	134,000	134,000	134,000	134,000	259,000
R-squared	0.39	0.67	0.44	0.70	0.74

Table 4

Cross-cluster innovation spillovers

This table presents variants of the regression in Eq. (2) in which the dependent variable is either the ratio (in logs) of the number of patents to the number of inventors (all in the plant's field) at the plant level or plant-level TFP. (Connected) cluster size is the total number of other firms' inventors (in the plant's field) across all other cities in which the firm has plants (in the plant's field) with inventors. Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Log(pat/inv) (1)	TFP (2)	Log(pat/inv) (3)	TFP (4)	Log(pat/inv) (5)	TFP (6)
Log(cluster size)	0.024*** (0.005)	0.010*** (0.003)	0.023*** (0.007)	0.012*** (0.004)	0.021*** (0.007)	0.012*** (0.004)
Field FE	Yes	Yes	–	–	–	–
Year FE	Yes	Yes	–	–	–	–
Field $\times$ Year FE	No	No	Yes	Yes	–	–
City $\times$ Year FE	No	No	Yes	Yes	–	–
City $\times$ Field FE	No	No	Yes	Yes	–	–
City $\times$ Field $\times$ Year FE	No	No	No	No	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	57,000	57,000	57,000	57,000	57,000	57,000
R-squared	0.45	0.69	0.52	0.75	0.59	0.79

4.4 Table 4: cross-cluster spillovers (Eq. 2)

Table 4 estimates Eq. (2).

- **Identification.** Columns (5)–(6) include $\text{city} \times \text{field} \times \text{year}$ FE, so estimates are not driven by any local city-field-year shock.
- **Main result.** Connected-cluster size has a positive and significant effect on both inventor productivity and TFPR.
- **Magnitude.** In the preferred specification (columns (5)–(6)), the elasticity is about 0.021 for inventor productivity and 0.012 for TFPR. Cross-cluster spillovers are smaller than within-cluster spillovers, but economically meaningful and precisely estimated.

4.5 Table 5: non-innovating plants and “placebo” connected clusters

Table 5 changes *which* nodes in the firm network are innovating.

- **Columns (1)–(2): innovating focal plants, but connected clusters contain only non-innovating plants.** Coefficients are near zero and insignificant. Interpretation: if a firm has no inventors in a connected city, there is no knowledge-producing node there, so there is little to diffuse back to the focal plant.
- **Columns (3)–(4): non-innovating focal plants.** When connected clusters contain innovating plants (column (3)), TFP rises significantly with connected-cluster size. When connected clusters contain only non-innovating plants (column (4)), the effect disappears. Interpretation: knowledge created by inventor nodes elsewhere in the firm can raise production efficiency even in plants without inventors, but only if the “source” locations are themselves innovating.

Table 5

Non-innovating plants.

This table presents variants of the regressions in columns (5) and (6) of Table 4. In columns (1) and (2), focal plants are innovating plants, and (connected) cluster size is the total number of other firms’ inventors (in the plant’s field) across all other cities where the firm has non-innovating plants. In columns (3) and (4), focal plants are non-innovating plants, fixed effects are based on the firm’s main field, and (connected) cluster size is the total number of other firms’ inventors (in the firm’s main field) across all other cities where the firm has innovating plants (in the firm’s main field) (column (3)) or non-innovating plants (column (4)). Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Log(pat/inv) (1)	TFP (2)	TFP (3)	TFP (4)
Log(cluster size)	0.002 (0.008)	0.001 (0.005)	0.006** (0.003)	0.002 (0.004)
City × Field × Year FE	Yes	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes
Observations	80,000	80,000	227,000	227,000
R-squared	0.52	0.75	0.82	0.82

Table 6

Instrumental variable estimation.

This table presents variants of the regressions in columns (5) and (6) of Table 4 estimated by 2SLS. The excluded instrument, Z , is the logarithm of the number of inventors (in the plant’s field) in other cities where the plant’s parent firm has no presence, and which are employed by other firms that have inventors in the plant’s connected clusters but not in the plant’s city. Column (1) shows the first stage; columns (2) and (3) shows 2SLS estimates where the main independent variable is predicted log(cluster size). Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Log(cluster size) (1)	Log(pat/inv) (2)	TFP (3)
Z	0.107*** (0.013)		
Log(cluster size)		0.023** (0.011)	0.014** (0.007)
City × Field × Year FE	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes
Observations	57,000	57,000	57,000
R-squared	0.85	0.59	0.79
F-statistic	70.60		

4.6 Table 6: instrumental-variable (2SLS) estimation

Table 6 addresses endogeneity concerns for cross-cluster spillovers by instrumenting connected-cluster size.

- **Instrument.** The excluded instrument Z is based on inventor counts in *other cities where the focal firm has no presence*, but where other firms (that are present in the focal plant’s connected clusters) employ inventors. Intuition: it shifts innovation conditions in connected clusters through other firms, while being plausibly orthogonal to focal-plant shocks (conditional on rich fixed effects).
- **First stage.** Z strongly predicts connected-cluster size (large F -statistic).
- **Second stage.** The 2SLS estimates remain positive and similar in magnitude to OLS. Interpretation: omitted variable bias from unobserved productivity shocks is unlikely to be the main driver of the OLS estimates.

4.7 Table 7: geographic distance and the lack of decay

Table 7 re-estimates the preferred cross-cluster specification after excluding connected clusters within 100, 250, or 500 miles of the focal plant. The coefficient remains stable and significant. Interpretation: within-firm spillovers need not rely on local face-to-face interactions and can propagate over long distances.

Table 7
Geographic distance.

This table presents variants of the regressions in columns (5) and (6) of Table 4 in which connected clusters within a 100, 250, or 500 mile radius around the plant are excluded. Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Log(pat./inv)			TFP		
	(1)	(2)	(3)	(4)	(5)	(6)
	Excluding clusters within 100 miles	Excluding clusters within 250 miles	Excluding clusters within 500 miles	Excluding clusters within 100 miles	Excluding clusters within 250 miles	Excluding clusters within 500 miles
Log(cluster size)	0.021*** (0.007)	0.020*** (0.007)	0.019** (0.008)	0.011*** (0.004)	0.011** (0.005)	0.011*** (0.005)
City × Field × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	53,000	46,000	35,000	53,000	46,000	35,000
R-squared	0.60	0.60	0.61	0.79	0.80	0.82

Table 8

Patent citations.

This table presents variants of the regressions in Table 4 and columns (1) and (2) of Table 5 in which the dependent variable is the share of the plant's citations to other firms' patents from connected clusters relative to all its citations to other firms' patents (in the plant's field). In column (4), citations to patents filed in year t are excluded. In column (5), (connected) cluster size is the total number of other firms' inventors (in the plant's field) across all other cities in which the firm has plants but no inventors. Observations are weighted by plant employment. Standard errors are double clustered at the city and year level. The sample period is from 1976 to 2018. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Patent citation share				
	(1)	(2)	(3)	(4)	(5)
Log(cluster size)	0.018** (0.007)	0.022*** (0.008)	0.019** (0.008)	0.019** (0.008)	0.002 (0.010)
Field FE	Yes	–	–	–	–
Year FE	Yes	–	–	–	–
Field × Year FE	No	Yes	–	–	–
City × Year FE	No	Yes	–	–	–
City × Field FE	No	Yes	–	–	–
City × Field × Year FE	No	No	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes	Yes
Observations	57,000	57,000	57,000	57,000	80,000
R-squared	0.55	0.63	0.69	0.69	0.58

4.8 Table 8: patent citations as a knowledge-flow mechanism

Table 8 uses the dependent variable

$$\text{PatentCitationShare}_{icft} = \frac{\text{citations to other firms' patents from connected clusters}}{\text{citations to other firms' patents (all locations)}},$$

computed within the plant's field.

- Connected-cluster size increases this citation share: plants draw more on knowledge from connected clusters when those clusters are larger.
- The effect survives excluding citations to patents filed in year t , mitigating purely contemporaneous-measurement concerns.
- The effect disappears for the placebo definition where “connected clusters” contain only non-innovating plants.

4.9 Tables 9–11: wedge dispersion and policy relevance

Table 9 (distribution and connectedness gradient). The wedge γ_n/β_n is computed for 179 BEA clusters, normalized so that the smallest wedge equals 1. The wedge displays dispersion around a mean above 1 (under-investment everywhere in this normalization). More connected clusters tend to have larger wedges; the correlation between wedges and connectedness is positive.

Table 9

Social-private innovation wedge

This table presents descriptive statistics for the social-private innovation wedge, γ_s/β_s . Wedges are standardized by scaling each wedge by the value of the smallest wedge, 1.109. In the right panel, tech clusters are sorted into deciles based on their average number of connected clusters across all plants in the cluster ("connectedness"). Column (4) shows the average wedge for each decile. Correlation refers to the correlation between the wedges and connectedness based on all 179 clusters.

Distribution of wedges (1)	(2)	Less vs. more connected clusters (3)	(4)
<i>Percentile</i>		<i>Decile</i>	
Min	1.000	1	1.086
5th	1.069	2	1.126
25th	1.123	3	1.141
50th	1.162	4	1.148
75th	1.195	5	1.165
95th	1.237	6	1.172
Max	1.269	7	1.167
		8	1.193
		9	1.197
Mean	1.156	10	1.165
Std. dev.	0.052		
N	179	Correlation: 0.411	

Table 10

Ranking U.S. tech clusters

This table ranks U.S. tech clusters based on their social-private innovation wedge, γ_s/β_s (left) and cluster size (right). Clusters are the 179 BEA economic areas ("cities"). Cluster size is the number of inventors in a city. Clusters are listed in ascending order.

Top 10 (social-private innovation wedge)	Top 10 (cluster size)
Atlanta-Sandy Springs-Gainesville, GA-AL	San Jose-San Francisco-Oakland, CA
Denver-Aurora-Boulder, CO	New York-Newark-Bridgeport, NY-NJ-CT-PA
Washington-Baltimore-Northern Virginia, DC	Minneapolis-St. Paul-St. Cloud, MN-WI
Los Angeles-Long Beach-Riverside, CA	Detroit-Warren-Flint, MI
San Jose-San Francisco-Oakland, CA	Boston-Worcester-Manchester, MA-NH
Philadelphia-Camden-Vineland, PA-NJ-DE-MD	Los Angeles-Long Beach-Riverside, CA
Chicago-Naperville-Michigan City, IL-IN-WI	Chicago-Naperville-Michigan City, IL-IN-WI
Phoenix-Mesa-Scottsdale, AZ	Portland-Vancouver-Seaswett, OR-WA
Raleigh-Durham-Cary, NC	Cleveland-Akron-Elyria, OH
Houston-Springtown-Humble, TX	
Bottom 10 (social-private innovation wedge)	Bottom 10 (cluster size)
Dodan-Enterprise-Ozark, AL	San Angelo, TX
Cape Girardeau-Jackson, MO-IL	Rapid City, SD
Farrington, NM	Providence-Rhode Island, RI
Providence-Rhode Island, RI	Albino, TX (0.4%)
Alpena, MI	Honolulu, HI
Bismarck, ND	Albino, TX
Milwaukee, WI	Aberdeen, SD
Rangor, ME	Minot, ND
Scotts Bluff, NE	Great Falls, MT
Great Falls, MT	Anchorage, AK
	Santa Fe-Spangula, NM

Table 11

Increasing the interconnectedness of U.S. tech clusters

This table ranks U.S. tech clusters based on the percent change in their social-private innovation wedge following a 10% increase in the off-diagonal elements of the S matrix (left) and cluster size (right). Clusters are the 179 BEA economic areas ("cities"). Cluster size is the number of inventors in a city. Clusters are listed in ascending order.

Top 10 (change in wedge)	Top 10 (cluster size)
Chicago-Naperville-Michigan City, IL-IN-WI (7.2%)	San Jose-San Francisco-Oakland, CA
Washington-Baltimore-Northern Virginia, DC (6.6%)	New York-Newark-Bridgeport, NY-NJ-CT-PA
Austin-Round Rock, TX (6.2%)	Minneapolis-St. Paul-St. Cloud, MN-WI
Boston-Worcester-Manchester, MA-NH (6.1%)	Detroit-Warren-Flint, MI
Indianapolis-Anderson-Columbus, IN (6.1%)	Boston-Worcester-Manchester, MA-NH
Houston-Springtown-Humble, TX (5.8%)	Los Angeles-Long Beach-Riverside, CA
San Diego-Carlsbad-San Marcos, CA (5.6%)	Chicago-Naperville-Michigan City, IL-IN-WI
Richmond-Bethesda-Silver Spring, MD (5.5%)	Portland-Vancouver-Seaswett, OR-WA
Seattle-Tacoma-Olympia, WA (5.3%)	Dallas-Fort Worth, TX
Pittsburgh-New Castle, PA (5.3%)	Cleveland-Akron-Elyria, OH
Bottom 10 (change in wedge)	Bottom 10 (cluster size)
Amarillo, TX (0.4%)	San Angelo, TX
Flagstaff, AZ (0.4%)	Rapid City, SD
Albino, TX (0.4%)	Providence-Rhode Island, RI
Kearney, NE (0.3%)	Honolulu, HI
Alpena, MI (0.2%)	Albino, TX
Gulfport-Biloxi-Pascagoula, MS (0.0%)	Aberdeen, SD
Juneau, AK (-0.0%)	Minot, ND
Fort Smith, AR-OK (-0.1%)	Great Falls, MT
Paducah, KY IL (-0.2%)	Anchorage, AK
Pueblo, CO (-0.2%)	Santa Fe-Spangula, NM

Table 10 (rankings by wedge vs. by size). The clusters with the largest wedges are *not* the same as the clusters with the most inventors. This is the core policy point: network position matters beyond local scale. Concretely, some very large clusters (by inventor count) are not the most under-investing ones, and some medium clusters rank very high in wedge.

Table 11 (counterfactual: increase interconnectedness). After increasing the off-diagonal elements of S by 10%, the percent change in the wedge varies widely across locations. The largest changes occur in large and well-connected clusters, consistent with the Leontief-network amplification logic in the theory.

5 Theoretical framework: model, welfare, and the innovation wedge

The model is designed to rationalize the reduced-form facts and to deliver a policy object that can be computed using sufficient statistics.

5.1 Setup and preferences (Eqs. (3)–(4))

There are N locations. Each location n has a representative consumer with preferences

$$U_n = \ln Y_n - L_n, \quad (3)$$

$$Y_n \equiv \left[\sum_i \int y_{ni}(\nu) \frac{\sigma-1}{\sigma} d\nu \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1. \quad (4)$$

The budget constraint is

$$\sum_i \int p_{ni}(\nu) y_{ni}(\nu) d\nu = w_n L_n + \Pi_n - T_n, \quad (5)$$

and wages are normalized to $w_n \equiv 1$.

5.2 CES demand, pricing, and profits (Eqs. (5)–(8))

CES demand implies:

$$p_{ni}(\nu) y_{ni}(\nu) = \frac{p_{ni}(\nu)^{1-\sigma}}{\sum_{i'} \int p_{ni'}(\nu')^{1-\sigma} d\nu'}. \quad (6)$$

Plants face iceberg trade costs τ_{ni} and have marginal cost $\kappa_i(\nu)$. With elasticity σ , monopolistic-competition pricing is:

$$p_{ni}(\nu) = \frac{\sigma}{\sigma - 1} \tau_{ni} \kappa_i(\nu). \quad (7)$$

Variable profits satisfy:

$$\pi_{ni}(\nu) = \frac{1}{\sigma} p_{ni}(\nu) y_{ni}(\nu). \quad (8)$$

5.3 Market access and plant revenues (Eqs. (6), (9), (13))

Define the **market access** term:

$$b_i \equiv \sum_n \frac{1}{\sigma} \frac{\tau_{ni}^{1-\sigma}}{\sum_{i'} \int (\tau_{ni'} \kappa_{i'}(\nu'))^{1-\sigma} d\nu'}. \quad (9)$$

With the normalization $\kappa_{nJ}^{1-\sigma} = k_{nJ}$, this can be re-expressed as:

$$b_i = \sum_n \frac{1}{\sigma} \frac{\tau_{ni}^{1-\sigma}}{\sum_m \sum_{J \ni m} N_J k_{mJ} \tau_{nm}^{1-\sigma}}. \quad (10)$$

Plant revenues satisfy $r_{nJ} = \sigma \pi_{nJ}$.

5.4 Plants, firms, and knowledge (Eqs. (10)–(11))

Fix a firm type J (the set of locations where the firm has plants), and denote by nJ the plant of firm J located in n . Each plant chooses innovation investment h_{nJ} . Plant knowledge is:

$$k_{nJ} \equiv \prod_{i \in J} \left(h_{iJ}^\alpha K_i^\delta \right)^{\omega_{ni}}, \quad (11)$$

where K_i is the location-wide knowledge stock in location i :

$$K_i \equiv \sum_{J \ni i} N_J k_{iJ}. \quad (12)$$

Parameters:

- $\alpha > 0$: returns to own innovation,
- $\delta > 0$: strength of *within-location* spillovers through K_i ,
- ω_{ni} : strength of *cross-location* spillovers from location i to plant in n .

Reduced-form guidance for ω_{ni} . The paper sets $\omega_{ni} = 1$ if $n = i$ and $\omega_{ni} = \omega < 1$ if $n \neq i$. When bringing the model to the data, ω_{ni} is set to zero if locations n and i are not connected through firms' plant networks.

5.5 Plant profits and the firm problem (Eqs. (12), (14)–(16))

Plant variable profits can be written as

$$\pi_{nJ} = b_n \prod_{i \in J} \left(h_{iJ}^\alpha K_i^\delta \right)^{\omega_{ni}}. \quad (13)$$

Given innovation subsidies $\{t_{nJ}\}$, firm J solves:

$$\hat{\pi}_J \equiv \max_{\{h_{nJ}\}} \sum_{n \in J} \left[b_n \prod_{i \in J} \left(h_{iJ}^\alpha K_i^\delta \right)^{\omega_{ni}} - \frac{c_n h_{nJ}}{t_{nJ}} \right]. \quad (14)$$

Aggregate profits are rebated:

$$\sum_n \Pi_n = \sum_J N_J \hat{\pi}_J, \quad (15)$$

and the government budget constraint is

$$\sum_n \left\{ \sum_{J \ni n} N_J \left(\frac{c_n h_{nJ}}{t_{nJ}} - c_n h_{nJ} \right) + T_n \right\} = 0. \quad (16)$$

Definition 1 (Equilibrium). *Given subsidies $\{t_{nJ}\}$, an equilibrium is a set of innovation choices $\{h_{nJ}\}$ and knowledge stocks $\{K_i\}$ such that: (i) firms solve (14); (ii) knowledge is consistent with (11)–(12); (iii) market access $\{b_i\}$ is consistent with (10); and (iv) goods and labor markets clear (the paper provides the explicit labor market clearing condition).*

5.6 Uniqueness (Lemma 1)

Lemma 1 (Uniqueness condition). *A sufficient condition for a unique equilibrium is*

$$(\alpha + \delta)(1 + (\mathcal{J} - 1)\omega) < 1, \quad (17)$$

where $\mathcal{J}$ is the maximum number of locations in which any firm operates (the maximum size of J).

Proof sketch (Appendix A.1). The equilibrium system defines a fixed-point mapping in the vectors $(\mathbf{h}, \mathbf{K})$. Using the mean value theorem, the difference between two candidate fixed points can be bounded by a Lipschitz constant equal to

$$(\alpha + \delta)(1 + (\mathcal{J} - 1)\omega).$$

If the constant is strictly smaller than one, the mapping is a contraction, implying uniqueness. $\square$

5.7 Welfare, private value, social value (Eqs. (18)–(24))

Social welfare is $\mathcal{W} \equiv \sum_n U_n$ and can be written as

$$\mathcal{W} = \sum_n \ln Y_n - L, \quad (18)$$

where $L \equiv \sum_n L_n$. The paper defines an alternative welfare object:

$$\widehat{\mathcal{W}} \equiv \sum_n \sum_{J \ni n} N_J \left(\frac{\sigma}{\sigma - 1} \pi_{nJ} - c_n h_{nJ} \right), \quad (19)$$

and shows (Lemma 2) that marginal welfare effects of innovation can be evaluated using $\widehat{\mathcal{W}}$ holding $\{b_n\}$ constant.

Plant-level private value. Define

$$\beta_{iJ} \equiv \frac{\partial \sum_{n \in J} \pi_{nJ}}{\partial \ln h_{iJ}} = \alpha \sum_{n \in J} \pi_{nJ} \omega_{ni} \mathbf{1}_{ni}^J. \quad (20)$$

The firm's FOC implies

$$\beta_{iJ} = \frac{c_i h_{iJ}}{t_{iJ}}. \quad (21)$$

Plant-level social value. Holding $\{b_n\}$ fixed, define

$$\gamma_{iJ} \equiv N_J^{-1} \frac{d\widehat{\mathcal{W}}}{d \ln h_{iJ}} \Big|_{\{b_n\}} \quad (22)$$

$$= \frac{\sigma}{\sigma - 1} N_J^{-1} \left(\alpha \sum_{n \in J} N_J \pi_{nJ} \omega_{ni} \mathbf{1}_{ni}^J + \delta \sum_{J'} N_{J'} \sum_{n \in J'} \sum_m \pi_{nJ'} \omega_{nm} \mathbf{1}_{nm}^{J'} \frac{d \ln K_m}{d \ln h_{iJ}} \right). \quad (23)$$

At the social optimum,

$$\gamma_{iJ} = c_i h_{iJ}. \quad (24)$$

Propagation of knowledge. Let

$$S_{nJ} \equiv \frac{r_{nJ}}{\sum_{J' \ni n} r_{nJ'}}$$

be plant nJ 's sales share of total sales in location n . Total differentiation yields the key recursion:

$$\frac{d \ln K_n}{d \ln h_{iJ}} = S_{nJ} \alpha \omega_{ni} \mathbf{1}_{ni}^J + \sum_{J' \ni n} S_{nJ'} \sum_{q \in J'} \delta \omega_{nq} \frac{d \ln K_q}{d \ln h_{iJ}}. \quad (25)$$

5.8 Optimal subsidy (Proposition 1) and special case (Eq. (26))

Comparing (21) and (24) yields the core identity:

$$t_{iJ} = \frac{\gamma_{iJ}}{\beta_{iJ}}.$$

Substituting the expressions above gives the paper's closed-form:

$$t_{iJ} = \frac{\sigma}{\sigma - 1} \frac{\alpha N_J \sum_{n \in J} r_{nJ} \omega_{ni} \mathbf{1}_{ni}^J + \delta \sum_{J'} N_{J'} \sum_{n \in J'} \sum_m r_{nJ'} \omega_{nm} \mathbf{1}_{nm}^{J'} \frac{d \ln K_m}{d \ln h_{iJ}}}{\alpha N_J \sum_{n \in J} r_{nJ} \omega_{ni} \mathbf{1}_{ni}^J}. \quad (26)$$

In the special case where there are no cross-location spillovers (or all firms are single-location), the optimal subsidy is uniform:

$$t = \frac{\sigma}{\sigma - 1} \cdot \frac{1}{1 - \delta}. \quad (27)$$

5.9 Location-level wedge (Proposition 2) and network interpretation

Let

$$s_{ni} \equiv \frac{\sum_J N_J r_{nJ} \mathbf{1}_{ni}^J}{\sum_J N_J r_{nJ}},$$

the revenue-weighted fraction of plants in location n that are connected to location i through firm networks. Let $S = [s_{ni}]$ and $\Omega = [\omega_{ni}]$; let $\mathbf{r}$ be total revenues by location.

Proposition 1 (Sufficient-statistic formulas). *Let $\boldsymbol{\beta} = [\beta_i]$ and $\boldsymbol{\gamma} = [\gamma_i]$ denote the vectors of (location-level) private and social innovation values. Then:*

1. $\boldsymbol{\beta}' = \frac{\alpha}{\sigma} \mathbf{r}' (\Omega \circ S)$,
2. $\boldsymbol{\gamma}' = \frac{\sigma}{\sigma - 1} \boldsymbol{\beta}' (I - \delta \Omega \circ S)^{-1}$,

where $\circ$ is the Hadamard product.

Proof outline (Appendix A.3). Aggregate (20) across plants/firms in a location and use $r = \sigma\pi$ to express $\boldsymbol{\beta}$ in terms of $\mathbf{r}$ and $(\Omega \circ S)$. Then use the linear system implied by (25) to express the total effect of innovation on location knowledge as a Leontief inverse, yielding the expression for $\boldsymbol{\gamma}$. $\square$

Leontief-network amplification.

$$(I - \delta \Omega \circ S)^{-1} = I + \delta \Omega \circ S + (\delta \Omega \circ S)^2 + \dots$$

so the wedge captures higher-order spillovers that propagate across the location network via multi-location firms.

Special-case wedge. In the special case behind (27), the location-level wedge is constant:

$$\frac{\gamma_n}{\beta_n} = \frac{\sigma}{\sigma - 1} \cdot \frac{1}{1 - \delta},$$

so dispersion in γ_n/β_n is entirely a *network phenomenon* in the general case.

6 Taking the wedge to the data: calibration logic behind Tables 9–11

To operationalize the wedge, the paper plugs reduced-form elasticities into the theoretical parameters:

- Within-cluster elasticity identifies δ (local spillovers).
- Cross-cluster elasticity identifies $\delta\omega$ (because cross-location spillovers enter through $\omega_{ni} = \omega$ off-diagonal).
- Hence ω is identified by the ratio $\omega = (\delta\omega)/\delta$.

Then, using observed connectedness matrices (built from firms’ plant networks) and revenue by location, the paper computes γ_n/β_n for each location and year and averages over time. Tables 9–11 summarize the dispersion, ranking, and counterfactual changes in this wedge.

7 Conclusion

7.1 Summary

This paper concludes that U.S. tech clusters generate innovation spillovers in *two* layers, and that the second layer—spillovers *across* clusters—operates primarily through **multi-location firms**.

The argument is one consistent chain:

1. **Within-cluster spillovers exist and are economically meaningful.** When the local city×field cluster becomes larger (more inventors at other firms), local innovating plants become more productive. In the preferred within-cluster specification (Table 3), the elasticity of plant-level inventor productivity with respect to local cluster size is about

$$\hat{\beta}^{\text{within}} \approx 0.076$$

and the elasticity of plant TFP is about

$$\hat{\beta}^{\text{within,TFP}} \approx 0.023.$$

A key placebo is that local cluster size has essentially no effect on TFP for plants *without* inventors (Table 3, column (5)), consistent with the idea that absorptive capacity matters.

2. **Cross-cluster spillovers also exist and can be identified cleanly.** The paper defines a plant’s *connected clusters* as other cities in the same field where the parent firm has innovating plants. Using within-plant variation in the size of these connected clusters plus city×field×year

fixed effects, the paper identifies cross-cluster spillovers (Table 4). In the preferred specification, the elasticity of inventor productivity with respect to connected-cluster size is about

$$\hat{\beta}^{\text{cross}} \approx 0.021,$$

with a corresponding TFP elasticity of about

$$\hat{\beta}^{\text{cross,TFP}} \approx 0.012.$$

3. **The cross-cluster channel is consistent with knowledge diffusion within firms.** Two mechanism facts support this interpretation:

- *No distance decay:* cross-cluster spillovers remain similar after excluding connected clusters within 100, 250, or 500 miles (Table 7).
- *Citation-based evidence:* as connected clusters become larger, plants cite a higher share of other-firm patents originating from those connected clusters (Table 8).

4. **The results survive an instrumental-variable design.** The paper instruments connected-cluster size with inventor activity in outside cities where the focal firm has no presence but where firms linked to the connected clusters have inventors. The first stage is strong (Table 6, $F \approx 70.6$), and 2SLS coefficients are similar in magnitude to OLS, suggesting that omitted productivity shocks are unlikely to be the main driver.

5. **Theory: spillovers imply systematic under-investment in innovation, and network position matters.** The paper develops a spatial innovation model with (i) within-location spillovers and (ii) cross-location spillovers mediated by firms' multi-location networks. It derives a sufficient statistic—the *social-private innovation wedge*—defined as

$$\frac{\gamma_n}{\beta_n},$$

where β_n is the private marginal value of innovation in location n and γ_n is the social marginal value. Calibrating the model using the within- and cross-cluster elasticities above, the paper computes wedges for 179 U.S. locations and shows meaningful dispersion (Table 9): the standardized wedge has mean about 1.156 and ranges roughly from 1.000 to 1.269.

6. **Policy implication: the highest-wedge clusters are not simply the largest clusters.** Wedges are more strongly related to connectedness than to size: across clusters, the correlation between wedges and connectedness is about 0.411, while the correlation with cluster size is about 0.166 (Tables 9–10). Hence, ranking clusters by wedge yields a different ordering than ranking by inventor counts (Table 10).

7.2 Economic intuition: why connected clusters generate distant spillovers

The intuition relies on combining a local spillover mechanism with the geography of firms.

1) Local spillovers create knowledge at the source. Within a city×field cluster, more nearby inventors at other firms increase the local stock of ideas and raise the productivity of local inventors. This is the standard “cluster externality” channel documented in Table 3.

2) Multi-location firms transmit local knowledge to other places. If a firm operates innovating plants in multiple clusters, local knowledge created in cluster A can be transmitted internally to cluster B through firm-specific channels (e.g., internal communication, shared routines, personnel mobility, codified knowledge repositories). This creates *cross-cluster* spillovers that do not require geographic proximity.

3) Why distance does not attenuate the effect. If diffusion occurs inside the firm, it is not constrained by the same face-to-face frictions as inter-firm diffusion. Therefore, once a location is connected through the firm’s innovating-plant network, the key state variable is the *size of the connected knowledge environment*, not physical distance. This explains the stability of coefficients when excluding nearby clusters (Table 7).

4) Why the wedge is larger in more connected clusters. Private innovation decisions internalize the firm’s own benefits, but not the benefits to other firms and other locations connected through the network. When a location is more connected, an innovation there spills over to a larger set of downstream nodes, raising social returns relative to private returns. In the model, this amplification operates through a network multiplier (a Leontief-inverse-type object), so “central” clusters have larger wedges and therefore larger under-investment incentives absent policy.

7.3 What to remember (for research design)

- Empirically, the paper provides a clean way to identify *cross-location* knowledge spillovers using connected-cluster definitions plus city×field×year fixed effects.
- Mechanism evidence (no distance decay + citation reweighting) is essential: it ties the reduced-form estimates to a within-firm diffusion channel rather than generic spatial correlation.
- Theoretically, the main transferable object is the *social-private innovation wedge* γ_n/β_n : it converts empirical elasticities and a network structure into a policy-relevant ranking of where innovation subsidies (or other innovation-enhancing policies) have the highest social payoff.

7.4 Takeaway

The paper shows that tech-cluster spillovers are not only local: because most inventors work for multi-location firms, growth in one cluster raises innovation and productivity in distant connected clusters via within-firm knowledge diffusion, and this networked spillover structure implies that the social return to innovation exceeds the private return most strongly in large and well-connected clusters (so network position matters at least as much as cluster size).